

EvoFlow - OD control

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August 2026

1 Dilution according Beer Lambert's Law

The optical density OD, or absorbance is given by Beer-Lambert's law as

$$A = \varepsilon \cdot l \cdot c \quad (1)$$

where:

- α = absorbance (OD)
- ε = molar absorption coefficient
- c = concentration of absorbing species
- l = optical path length

stating

$$A \propto c \quad (2)$$

In the control system, ε and l are constant.

For a dilution with v_1 of solution and v_2 as solvent with $A = 0$ a new concentration c_{new} can be calculated by

$$c_{new} = c_{old} \frac{v_1}{v_1 + v_2} \quad (3)$$

and therefore

$$A_{new} = A_{old} \frac{v_1}{v_1 + v_2} \quad (4)$$

2 Dynamic model

After priming the system, the volume V in the bioreactor is constant, meaning, the inflow into the vessel q_{in} is equal to the outflow q_{out} . This is leaving us with the differential equation

$$\frac{dA}{dt} = -\frac{q}{V} A(t) \quad (5)$$

The negative sign comes from the decrease of A , when diluting the liquid.

The relevant control variable is then the dilution rate

$$D = \frac{q}{V} \quad (6)$$

with unit 1/s. This leads to the form

$$\dot{A}(t) = -DA(t) \quad (7)$$

for constant flow q , the solution of this ODE is

$$A(t) = A_0 e^{-Dt} \quad (8)$$

This exponential form states, that with no cell growth and a constant flow, the OD A decays to 0 for $t \rightarrow \infty$

3 Cell growth

Having live bacteria in the liquid, includes a disturbance. With

- $A \propto$ bacteria concentration
- constant vessel volume V
- inflow with absorbance $A = 0$
- inflow q_{in} equal to outflow q_{out}
- cell growth rate μ proportional to bacteria concentration or absorbance A with unit 1/s

the dynamics can be written as

$$\dot{A}(t) = \mu A(t) - DA(t) = (\mu - D)A(t) \quad (9)$$

At steady state at a controlled setpoint $A = A_{set}$ to remain constant, then

$$\dot{A} = 0 \quad (10)$$

or

$$(\mu - D)A = 0 \quad (11)$$

For a nonzero OD A this means

$$D = \mu \quad (12)$$

or equivalently

$$q = V\mu \quad (13)$$

This states, that keeping the OD constant, the flow q is primarily driven by the cell grow rate and not directly by the desired OD.

Having this error term, a feedforward term can be introduced:

$$D = \hat{\mu} + u_{PI} \quad (14)$$

where $\hat{\mu}$ is an estimate of the bacterial growth.

Having an error

$$e = A_{set} - A \quad (15)$$

the controller is then given by

$$D(t) = \hat{\mu} + k_p e + k_i \int e(t) dt \quad (16)$$

With

$$q = VD \quad (17)$$

this translates to

$$q(t) = V \left(\hat{\mu} + k_p e(t) + k_i \int e(t) dt \right) \quad (18)$$

As $q > 0$ reduces the OD, $k_p, k_i < 0$

4 Estimating the cell growth rate

The sampled solution of the differential equation for the dilution including bacterial control can be written as

$$A_{k+1} = A_k e^{(\mu_k - D_k)T_s} \quad (19)$$

Solving for μ_k leads to

$$\hat{\mu}_k = D_k + \frac{1}{T_s} \ln \left(\frac{A_{k+1}}{A_k} \right) \quad (20)$$

The growth rate can therefore be estimated by the sensor and control signals with

$$\hat{\mu}_k = \frac{q_{k-1}}{V} + \frac{1}{T_s} \ln \left(\frac{A_k}{A_{k-1}} \right) \quad (21)$$

5 Implementation and Limitations

For now, the flows were only written as q_{in} and q_{out} . However,

$$q_{out} = q_{out,1} + q_{out,2} \quad (22)$$

where $q_{out,1}$ is the flow to the lagoon and $q_{out,2}$ is the flow to the waste. As the flow to the lagoon $q_{out,1}(t) > 0$. To reach the condition

$$q_{in} = q_{out} \quad (23)$$

the flow to the wast $q_{out,2}$ is given by

$$q_{out,2} = q_{in} - q_{out,1} \quad (24)$$

For the current prototype the waste pump is controlled as

$$q_{out,2} = 1.1 \cdot q_{in} \quad (25)$$

to prevent spillage. Pumping too much liquid out of the bioreactor is prevented by setting the needle height of the waste line. As the flow to the lagoon in the running system is

$$q_{out,1} > 0 \quad (26)$$

and the needle is submerged

$$q_{out} = 1.1 \cdot q_{in} + q_{out,1} \quad (27)$$

would change the volume V in the bioreactor with $q_{out,2} \cdot t$. This would break the model assumption V const. and would lead to inaccuracies in the estimation of bacterial growth and therefor in the control loop. To prevent spillage or overflow, a cyclic "overpumping" of the wasteline can be implemented, to compensate for model errors.

As outflow q_{out} of the bioreactor needs to be compensated by inflow q_{in} , and there is always a flow $q_{out,1}$ to the lagoon that needs to be maximized during the experiment, this flow to the lagoon acts as a disturbance to the OD controller, that can not be actively compensated by the OD controller. As the OD controller can only actively decrease OD and only bacterial growth increases OD, the flow to the lagoon, can only reach a certain level.

6 Some numerical magic

Maximal flow to keep OD constant

The bacterial reproduction takes approximately

$$\tau \approx 45 \text{ min} = 2.700\text{e}3 \text{ s} \quad (28)$$

The solution for the differential equation including bacterial growth is

$$A(t) = A_0 e^{(\mu-D)t} \quad (29)$$

The bacterial growth rate μ can be calculated by reformulating the equation

$$2A_0 = A_0 e^{(\mu-D)\tau} \quad (30)$$

for μ

$$\mu = \frac{\ln(2)}{\tau} + D = \frac{\ln(2)}{\tau} + \frac{q}{V} \quad (31)$$

having no inflow $q = 0$ leads to

$$\mu \approx \frac{\ln(2)}{2700 \text{ s}} \approx \underline{\underline{260\text{e-}6 \text{ s}^{-1}}} \quad (32)$$

having no change in OD

$$\dot{A}(t) = 0 \quad (33)$$

leads to

$$q_{const} = V\mu \approx 30 \text{ mL} \cdot 260\text{e-}6 \text{ s}^{-1} \approx 7.7\text{e-}3 \text{ mL s}^{-1} = 27.7 \text{ mL h}^{-1} \quad (34)$$